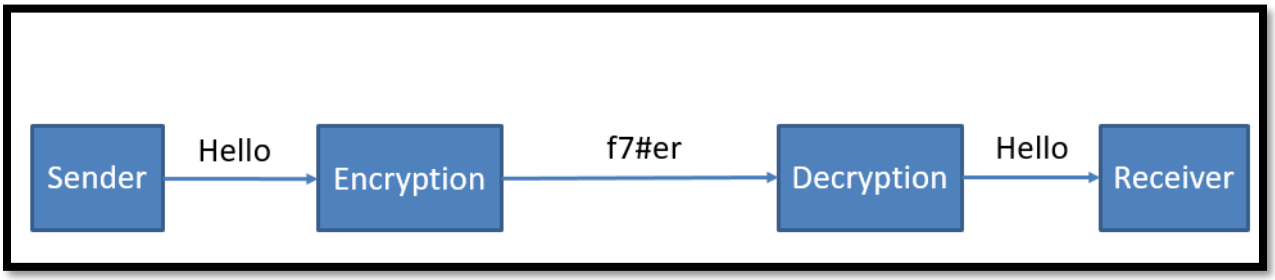
Unit: 2 Classical Encryption Techniques

Topics:

1. Symmetric Cipher Model
2. Substitution Techniques
3. Transposition Techniques
4. Steganography

2. Introduction



Encryption and Decryption [2]

Terminologies

Plaintext: original message

Cipher text: encrypted or coded message

Encryption: convert from plaintext to cipher text (enciphering)

Decryption: restore the plaintext from cipher text (deciphering)

Key: information used in cipher known only to sender/receiver

Cipher: a particular algorithm (cryptographic system)

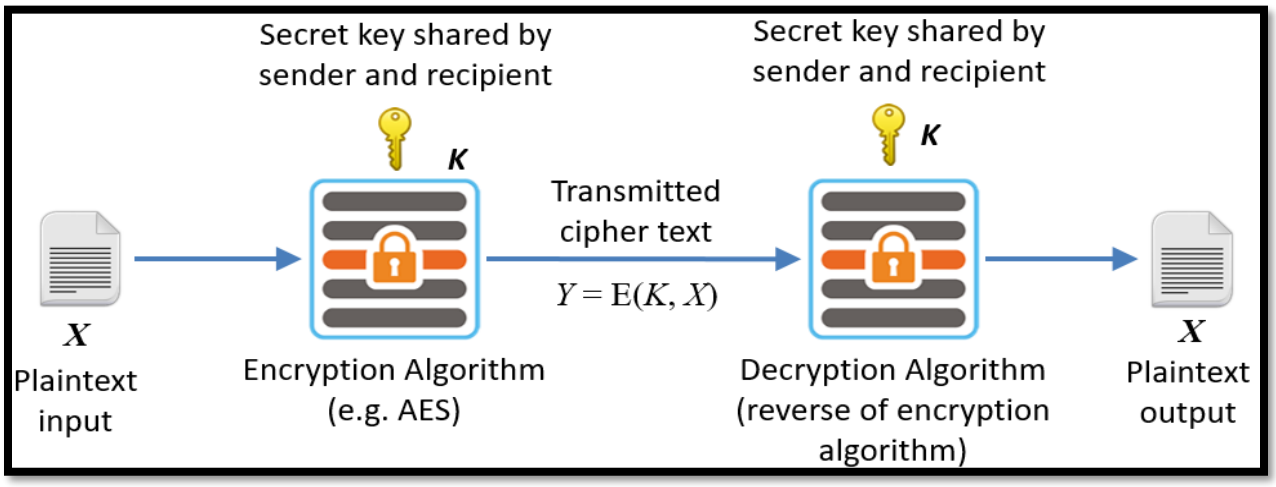
Cryptography: study of algorithms used for encryption

Cryptanalysis: study of techniques used for decryption without knowledge of plaintext

Cryptology: areas of cryptography and cryptanalysis

2.1 Symmetric Cipher Model

- An original message is known as the **plaintext**, while the coded message is called the **cipher text**.



Symmetric Cipher Model [4]

- The process of converting from plaintext to cipher text is known as **enciphering** or **encryption**; restoring the plaintext from the cipher text is **deciphering** or **decryption**.
- **Plaintext** is the original intelligible message or data that is fed into the algorithm as input.
- **Encryption algorithm** performs various substitutions and transformations on the plaintext.
- **Decryption algorithm** is essentially the encryption algorithm run in reverse.
- It takes the cipher text and the secret key and produces the original plaintext.
- Two requirements for secure use of conventional encryption.

1. We need a strong encryption algorithm. At a minimum, we would like the algorithm to be such that an opponent who knows the algorithm and has access to one or more cipher texts would be unable to decipher the cipher text or figure out the key.

    - This requirement is usually stated in a strong form: The opponent should be unable to decrypt cipher text or discover the key even if he or she is in possession of a number of cipher texts together with the plaintext that produced each cipher text.

2. Sender and receiver must have obtained copies of the secret key in a secure fashion and must keep the key secure. If someone can discover the key and knows the algorithm, all communication using this key is readable.

Plain text X=[x1, x2, x3...]

Key K= [k1, k2...]

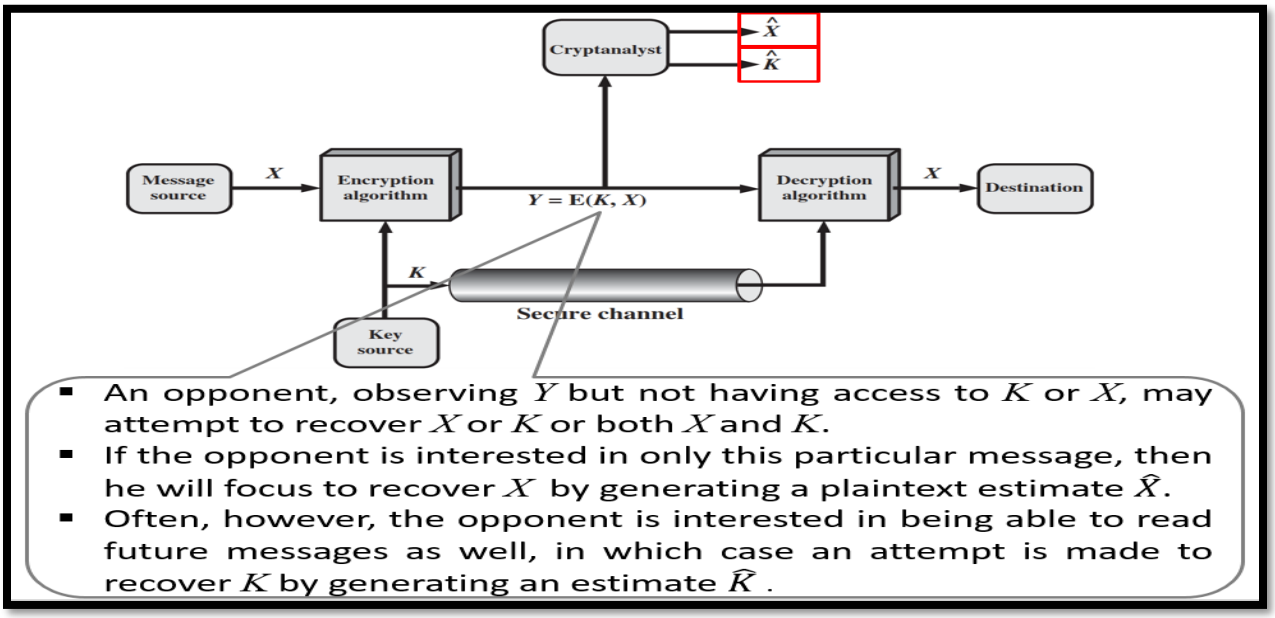
Cipher text Y= [y1, y2...]

Encryption

Y=E (K, X)

Decryption

X=D (K, Y)



Symmetric Cipher Model [3]



Cryptography and Cryptanalysis

- Cryptography (Secret Writing) is the process of protecting information by transforming it into a secure (unreadable) format.



Cryptography [2]

- Cryptanalysis is the decryption and analysis of encrypted text. Cryptanalysis uses mathematical formulas to search algorithm vulnerabilities and break into cryptography.



Cryptanalysis [2]

Cryptography

- Cryptography is the study of mathematical techniques related to aspects of information security such as **confidentiality**, **data integrity**, and entity/data origin **authentication**.

- Characterized in three dimensions:

  Type of operation used

  1) Substitution : Elements of plaintext are mapped into another element

  2) Transposition: Elements of plaintext are rearranged

- The Number of keys used

  1) Symmetric, Single-key, Secret-key, Conventional Encryption: Same Key is Used

  2) Asymmetric, two-key, Public-key: different keys are used



Way in which the plaintext is processed

1) Stream Cipher: processes the input elements continuously, producing output **one element at a time**.

2) Block Cipher: processes the input **one block** of elements at a time, producing an **output block** for each input block.

Cryptanalysis

- **Cryptanalysis** is the study of mathematical techniques for attempting to defeat cryptographic techniques that provide information security services.
- Cryptanalysis means "**breaking the code**".
- Techniques used for deciphering a message without any knowledge of enciphering details fall into the area of cryptanalysis
- Typically, the objective of attacking an encryption system is to **recover the key** in use rather than simply to **recover the plain text.**
- Two approaches:
  - cryptanalytic attack
  - brute-force attack
- **Brute-force attack**

  A brute-force attack involve trying every possible key until an intelligible translation of the cipher text into plaintext is obtained.



Key Size (bits)	Number of Alternative Keys	Time required at 1 decryption/μs	Time required at 10^6 decryptions/μs
32	$2^{32} = 4.3 \times 10^9$	2^{31} μs = 35.8 minutes	2.15 milliseconds
56	$2^{56} = 7.2 \times 10^{16}$	2^{55} μs = 1142 years	10.01 hours
128	$2^{128} = 3.4 \times 10^{38}$	2^{127} μs = 5.4×10^{24} years	5.4×10^{18} years
168	$2^{168} = 3.7 \times 10^{50}$	2^{167} μs = 5.9×10^{36} years	5.9×10^{30} years
26 characters (permutation)	$26! = 4 \times 10^{26}$	2×10^{26} μs = 6.4×10^{12} years	6.4×10^6 years

Brute force attack [2]

Cryptanalysis attacks

- Cryptanalysis attacks rely on
  - the nature of the algorithm plus
  - Some knowledge of the general characteristics of the plain text or even some sample plaintext-cipher text pairs.

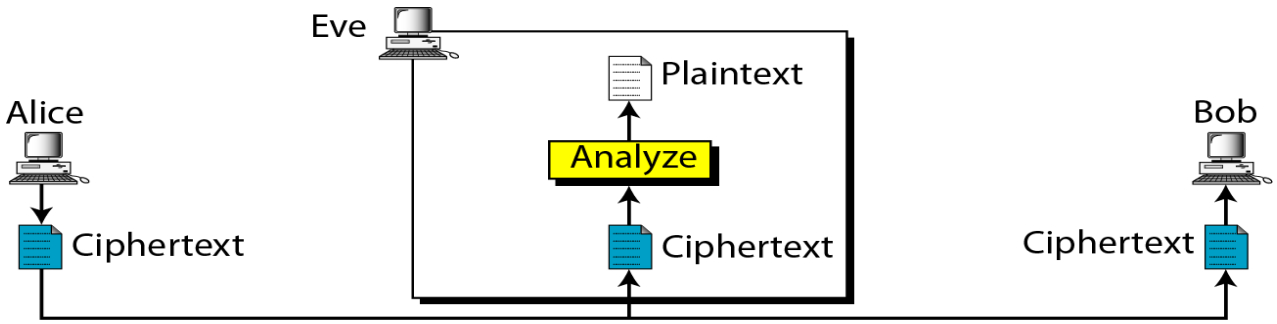
Type of Attack	Known to Cryptanalyst
Ciphertext Only	• Encryption algorithm • Ciphertext
Known Plaintext	• Encryption algorithm • Ciphertext • One or more plaintext–ciphertext pairs formed with the secret key
Chosen Plaintext	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key
Chosen Ciphertext	• Encryption algorithm • Ciphertext • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key
Chosen Text	• Encryption algorithm • Ciphertext • Plaintext message chosen by cryptanalyst, together with its corresponding ciphertext generated with the secret key • Ciphertext chosen by cryptanalyst, together with its corresponding decrypted plaintext generated with the secret key

Types of Attacks [3]

- This type of attack exploits the characteristics of the algorithm to attempt to **deduce a specific plaintext** or to **deduce the key** being used.

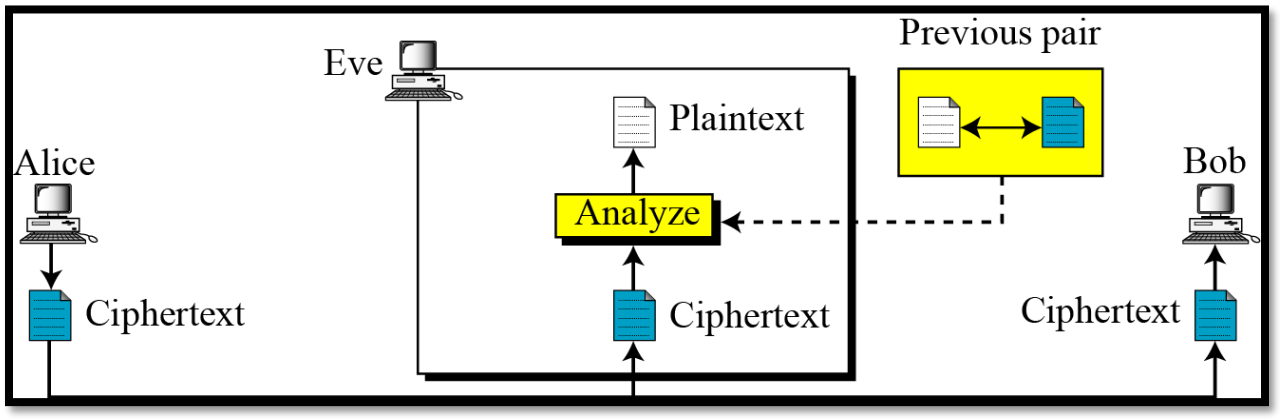
Attacks on Encrypted Messages

Type of Attack	Known to cryptanalyst
Cipher text Only	Encryption algorithm, Cipher text



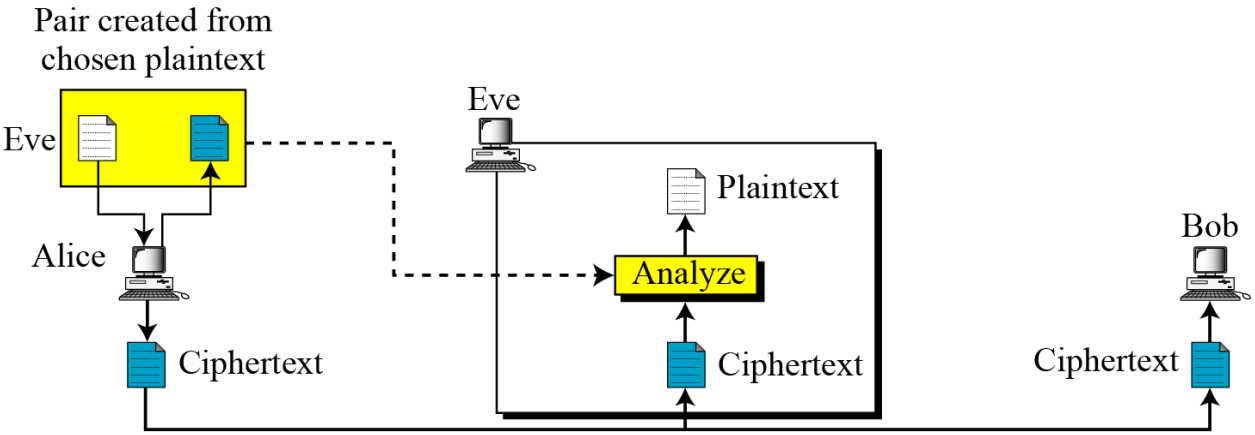
Cipher text only attack [2]

Type of Attack	Known to cryptanalyst
Known Plaintext	Encryption algorithm, Cipher text, One or more plaintext-cipher text pairs formed with the secret key



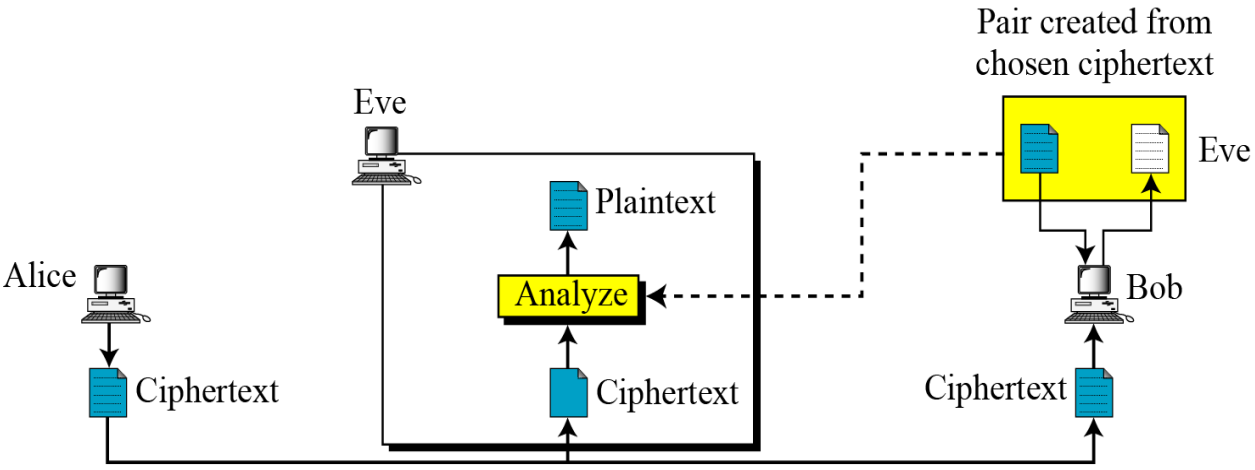
Attacks on Encrypted Messages [3]

Type of Attack	Known to cryptanalyst
Chosen Plaintext	Encryption algorithm, Cipher text, Plaintext message chosen by cryptanalyst



Attacks on Encrypted Messages [3]

Type of Attack	Known to cryptanalyst
Chosen Cipher text	Encryption algorithm, Cipher text, Cipher text chosen by cryptanalyst, with its corresponding decrypted plaintext generated with the secret key



Attacks on Encrypted Messages [3]



Type of Attack	Known to cryptanalyst
Chosen text	Encryption algorithm, Cipher text, Plaintext chosen by cryptanalyst, with its corresponding cipher text generated with the secret key , Cipher text chosen by cryptanalyst, with its corresponding decrypted plaintext generated with the secret key

Unconditional Security

- An encryption scheme is unconditionally secure if the cipher text generated by the scheme does not contain enough information to determine uniquely the corresponding plaintext, no matter how much cipher text is available.

- That is, no matter how much computer power or time is available, the cipher cannot be broken since required information is not there.

- There is no encryption algorithm that is unconditionally secure except one-time pad. So,
    - The cost of breaking cipher exceeds the value of encrypted information
    - The time required to break cipher exceeds the useful lifetime of the information

Computational Security

- An encryption scheme is said to be computationally secure if either of the following two criteria are met.

- The cost of breaking the cipher exceeds the values of the encrypted information.

- The time required to break the cipher exceeds the useful lifetime of the information.

Symmetric key cryptography

- An original message is known as the plaintext, while the coded message is called the cipher text.



- The process of converting from plaintext to cipher text is known as encryption or enciphering.
- Restoring the plaintext from the cipher text is known as decryption or deciphering.

- Techniques used for deciphering a message without any knowledge of the enciphering details fall into the area of cryptanalysis.

- Symmetric encryption is a form of cryptosystem in which encryption and decryption are performed using the same key. It is also known as conventional encryption.

- Symmetric encryption transform plaintext into cipher text using a secret key and an encryption algorithm.

- Using the same key and a decryption algorithm, the plaintext is recovered from the cipher text.

Classical Encryption Techniques

- The two basic building blocks of all encryption techniques are

- Substitution Techniques

    - In this the letters of plaintext are replaced by other letters or by numbers or symbols

- Transposition Techniques

    - Performs some sort of permutation on the plaintext letters

    - Rearrangement of plaintext to produce cipher text



2.2 Substitution Techniques

- A substitution technique is one in which the letters of plaintext are replaced by other letters or by numbers or symbols.

1) Caesar Cipher
2) Mono alphabetic Cipher
3) Play fair Cipher
4) Hill Cipher
5) Polyalphabetic Ciphers
6) One-Time Pad

1) Caesar Cipher

- The Caesar cipher involves replacing each letter of the alphabet with the letter standing three places further down the alphabet.

- In encryption each plaintext letter P, substitute the cipher text letter C:

$$C = E(k, P) = (P + k) \bmod 26$$
$$C = E(3, P) = (P + 3) \bmod 26$$

For decryption algorithm is:

$$P = D(k, C) = (C - k) \bmod 26$$

- Let us assign a numerical equivalent to each letter

$$C = E(3, P) = (P + 3) \bmod 26$$



a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

plain: a b c d e f g h i j k l m n o p q r s t u v w x y z

cipher: d e f g h i j k l m n o p q r s t u v w x y z a b c

Example:

Plain text: T H E Q U I C K B R O W N F O X
Cipher text: W K H T X L F N E U R Z Q I R A

Brute force attack on Caesar Cipher

- The encryption and decryption algorithms are known.
- There are only 25 keys to try.
- The language of the plaintext is known and easily recognizable.

Cipher text: ZNK WAOIQ HXUCT LUD

Key	Transformed text
1	YMJ VZNHP GWTBS KTC
2	XLI UYMGO FVSAR JSB
3	WKH TXLFN EURZQ IRA
4	VJG SWKEM DTQYP HQZ
5	UIF RVJDL CSPXOGPY
6	THE QUICK BROWN FOX
7	SGD PTHBJ AQNVM ENW
8	RFC OSGAI ZPMUL DMV
9	QEB NRFZH YOLTK CLU
10	PDA MQEYG XNKSJ BKT
11	OCZ LPDXF WMJRI AJS
12	NBY KOCWE VLIQH ZIR
13	MAX JNBVD UKHPG YHQ

Brute force attack on Caesar cipher [3]

Key	Transformed text
14	LZW IMAUC TJGOF XGP
15	KYV HLZTB SIFNE WFO
16	JXU GKYSA RHEMD VEN
17	IWT FJXRZ QGDLC UDM
18	HVS EIWQY PFCKB TCL
19	GUR DHVPX OEBJA SBK
20	FTQ CGUOW NDAIZ RAJ
21	ESP BFTNV MCZHY QZI
22	DRO AESMU LBYGX PYH
23	CQN ZDRLT KAXFW OXG
24	BPM YCQKS JZWEV NWF
25	AOL XBPJR IYVDU MVE

Brute force attack on Caesar cipher [3]



2) Monoalphabetic Cipher (Simple substitution)

- It is an improvement to the Caesar Cipher.
- Instead of shifting the alphabets by some number, this scheme uses some permutation of the letters in alphabet.

  plain: a b c d e f g h i j k l m n o p q r s t u v w x y z
  cipher: y n l k x b s h m i w d p j r o q v f e a u g t z c

- The sender and the receiver decide on a randomly selected permutation of the letters of the alphabet.
- With 26 letters in alphabet, the possible permutations are 26! which is equal to 4x1026.

Attack on Monoalphabetic Cipher

The relative frequencies of the letters in the cipher text (in percentages) are

P 13.33	H 5.83	F 3.33	B 1.67	C 0.00
Z 11.67	D 5.00	W 3.33	G 1.67	K 0.00
S 8.33	E 5.00	Q 2.50	Y 1.67	L 0.00
U 8.33	V 4.17	T 2.50	I 0.83	N 0.00
O 7.50	X 4.17	A 1.67	J 0.83	R 0.00
M 6.67				

Attack on Mono alphabetic cipher [3]

Cipher text:

Uzqsovuohxmopvgpozpevsgzwszopfpesxudbmetsxaizvuephzhmdzshzowsfpappdtsvpq
uzwymxuzuhsxepyepopdzszufpombzwpfupzhmdjudtmohmq

- In our cipher text, the most common digram is ZW, which appears three times. So equate Z with t, W with h and P with e.
- Now notice that the sequence ZWP appears in the ciphertext, and we can translate that sequence as “the.”
- If the cryptanalyst knows the nature of the plaintext, then the analyst can exploit the regularities of the language.
- The relative frequency of the letters can be determined and compared to a standard frequency distribution for English.



- If the message were long enough, this technique alone might be sufficient, but because this is a relatively short message, we cannot expect an exact match.

3) Playfair Cipher

The Playfair algorithm is based on a 5 × 5 matrix (key) of letters.
The matrix is constructed by filling in the letters of the keyword (minus duplicates) from left to right and from top to bottom, and then filling in the remainder of the matrix with the remaining letters in alphabetic order. The letters I and J count as one letter.

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z

Play fair cipher [3]

Example:
Keyword= OCCURRENCE
Plaintext= TALL TREES

- Playfair, treats digrams (two letters) in the plaintext as single units and translates these units into cipher text digrams.

- Make Pairs of letters add filler letter “X” if same letter appears in a pair.

  Plaintext= TALL TREES
  Plaintext= TA LX LT RE ES

- If there is an odd number of letters, then add uncommon letter to complete digram, a X/Z may be added to the last letter.

- Playfair, treats digrams (two letters) in the plaintext as single units and translates these units into cipher text digrams.

- Make Pairs of letters add filler letter “X” if same letter appears in a pair.



Plaintext= TALL TREES

Plaintext= TA LX LT RE ES

- If there is an odd number of letters, then add uncommon letter to complete digram, an X/Z may be added to the last letter.

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z

Play fair cipher [3]

- For example, using the table above, the letter pair TA would be encoded as PF.

- Plaintext= TA LX LT RE ESthe letters appear on the same column, replace them with the letters immediately below, wrapping around to the top if necessary.

- For example, using the table above, the letter pair LT would be encoded as TZ.

1. Key= " engineering " Plaintext=" test this process "
2. Key= " keyword " Plaintext=" come to the window "
3. Key= " moon mission " Plaintext=" greet "



E N G I R A B C D F H K L M O P Q S T U V W X Y Z	Encrypted Message: pi tu pm gt ue lf gp xg

K E Y W O R D A B C F G H I L M N P Q S T U V X Z	Encrypted Message: lc nk zk vf yo gq ce bw

M O N I S A B C D E F G H K L P Q R T U V W X Y Z	Encrypted Message: hq cz du

Play fair cipher [3]

4) Hill Cipher

- Hill cipher is based on linear algebra
- Each letter is represented by numbers from 0 to 25 and calculations are done modulo 26.
- Encryption and decryption can be given by the following formula:

  Encryption:

$$C = PK \bmod 26$$



Decryption:

$$P=CK^{-1} \bmod 26$$

$$(c_1\ c_2\ c_3) = (p_1\ p_2\ p_3) \begin{pmatrix} k11 & k12 & k13 \\ k21 & k22 & k23 \\ k31 & k32 & k33 \end{pmatrix} \bmod 26$$

- To encrypt a message using the Hill Cipher we must first turn our keyword and plaintext into a matrix (a 2 x 2 matrix or a 3 x 3 matrix, etc).

  Example: Key = “HILL”, Plaintext = “EXAM”

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12
n	**o**	**p**	**q**	**r**	**s**	**t**	**u**	**v**	**w**	**x**	**y**	**z**
13	14	15	16	17	18	19	20	21	22	23	24	25

$$\text{Key Matrix} \begin{pmatrix} H & I \\ L & L \end{pmatrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}$$

$$\text{Plaintext} \begin{pmatrix} E \\ X \end{pmatrix} \begin{pmatrix} A \\ M \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix}$$



$$C = PK \bmod 26$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}\begin{pmatrix} 4 \\ 23 \end{pmatrix}$$

$$7 \times 4 + 8 \times 23 = 212$$

$$11 \times 4 + 11 \times 23 = 297$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}\begin{pmatrix} 4 \\ 23 \end{pmatrix} = \begin{pmatrix} 212 \\ 297 \end{pmatrix}$$

$$\begin{pmatrix} 212 \\ 297 \end{pmatrix} = \begin{pmatrix} 4 \\ 11 \end{pmatrix} \bmod 26 = \begin{pmatrix} E \\ L \end{pmatrix}$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}\begin{pmatrix} 0 \\ 12 \end{pmatrix}$$

$$7 \times 0 + 8 \times 12 = 96$$

$$11 \times 0 + 11 \times 12 = 132$$

$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}\begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 96 \\ 132 \end{pmatrix}$$

$$\begin{pmatrix} 96 \\ 132 \end{pmatrix} = \begin{pmatrix} 18 \\ 2 \end{pmatrix} \bmod 26 = \begin{pmatrix} S \\ C \end{pmatrix}$$

Ciphertext = “ELSC”

$$P = CK^{-1} \bmod 26$$

Step: 1 Find Inverse of key matrix

Step: 2 Multiply the Multiplicative Inverse of the Determinant by the Adjoin Matrix

Step: 3 Multiply inverse key matrix with cipher text matrix to obtain plaintext matrix

Step: 1 Inverse of key matrix

2 X 2 inverse of matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - cb}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$



3 X 3 inverse of matrix

$$A^{-1} = \frac{1}{\text{determinant}(A)} \cdot \text{adjoin}(A)$$

$$\text{Inverse Key Matrix} = \begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix}^{-1} = \frac{1}{77-88}\begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{-11}\begin{pmatrix} 11 & -8 \\ -11 & 7 \end{pmatrix}$$

$$= \frac{1}{15}\begin{pmatrix} 11 & 18 \\ 15 & 7 \end{pmatrix} \bmod 26$$

- -11 mod 26 = 15
- Because, modulo for negative number is = N- (B%N)
  = 26 – (11%26)

Step: 2 Modular (Multiplicative) Inverse

- The inverse of a number A is 1/A since A * 1/A = 1

e.g. the inverse of 5 is 1/5

- In modular arithmetic we do not have a division operation.
- The modular inverse of A (mod C) is A^{-1}
- $(A * A^{-1}) \equiv 1 \pmod{C}$

Example:

- The modular inverse of A mod C is the A^{-1} value that makes

$A * A^{-1} \bmod C = 1$

A = 3, C = 11

Since (3*4) mod 11 = 1, 4 is modulo inverse of 3

A = 10, C = 17 , A^{-1} = ? 12



Determinants' multiplicative inverse Modulo 26												
Determinant	**1**	**3**	**5**	**7**	**9**	**11**	**15**	**17**	**19**	**21**	**23**	**25**
Inverse Modulo 26	**1**	**9**	**21**	**15**	**3**	**19**	**7**	**23**	**11**	**5**	**17**	**25**

$$= \frac{1}{15}\begin{pmatrix}11 & 18\\ 15 & 7\end{pmatrix} \bmod 26$$

- Multiplicative inverse of $\frac{1}{15}$ is 7

Step: 3 Multiply with adjoin of matrix

$$= 7\begin{pmatrix}11 & 18\\ 15 & 7\end{pmatrix} = \begin{pmatrix}77 & 126\\ 105 & 49\end{pmatrix} = \begin{pmatrix}25 & 22\\ 1 & 23\end{pmatrix} \bmod 26$$

$$= \text{thus, if } K = \begin{pmatrix}7 & 8\\ 11 & 11\end{pmatrix} \text{ then } K^{-1} = \begin{pmatrix}25 & 22\\ 1 & 23\end{pmatrix}$$

Hill Cipher Encryption

$$\text{Inverse Key Matrix} = \begin{pmatrix}25 & 22\\ 1 & 23\end{pmatrix}$$

$$\text{Ciphertext} \begin{pmatrix}E\\ L\end{pmatrix}\begin{pmatrix}S\\ C\end{pmatrix} = \begin{pmatrix}4\\ 11\end{pmatrix}\begin{pmatrix}18\\ 2\end{pmatrix}$$

$$P = CK^{-1} \bmod 26$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix}$$

$$25 \times 4 + 22 \times 11 = 342$$

$$1 \times 4 + 23 \times 11 = 257$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 4 \\ 11 \end{pmatrix} = \begin{pmatrix} 342 \\ 257 \end{pmatrix}$$

$$\begin{pmatrix} 342 \\ 257 \end{pmatrix} = \begin{pmatrix} 4 \\ 23 \end{pmatrix} \bmod 26 = \begin{pmatrix} E \\ X \end{pmatrix}$$

$$\begin{pmatrix} 25 & 22 \\ 1 & 23 \end{pmatrix} \begin{pmatrix} 18 \\ 2 \end{pmatrix}$$

$$25 \times 18 + 22 \times 2 = 494$$

$$1 \times 18 + 23 \times 2 = 64$$

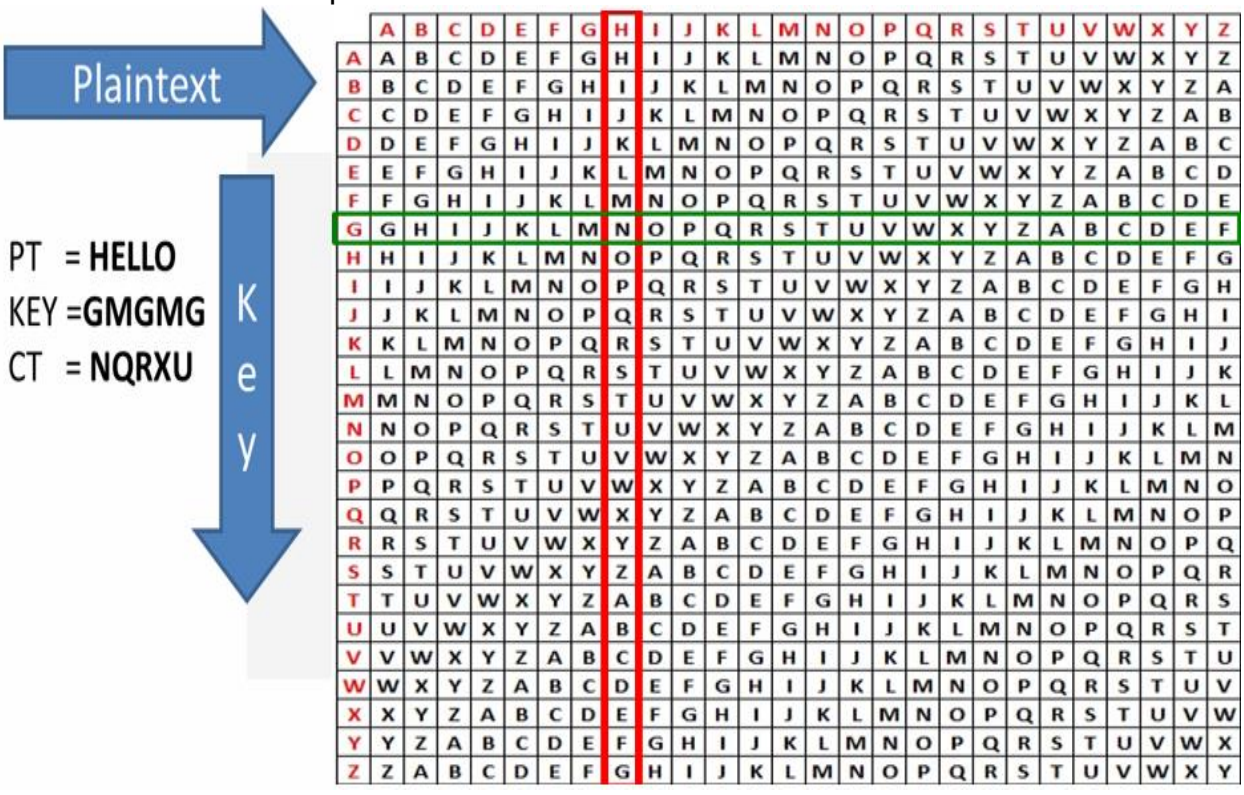
$$\begin{pmatrix} 7 & 8 \\ 11 & 11 \end{pmatrix} \begin{pmatrix} 0 \\ 12 \end{pmatrix} = \begin{pmatrix} 494 \\ 64 \end{pmatrix}$$

$$\begin{pmatrix} 494 \\ 64 \end{pmatrix} = \begin{pmatrix} 0 \\ 12 \end{pmatrix} \bmod 26 = \begin{pmatrix} A \\ M \end{pmatrix}$$

Plaintext = “EXAM”

5) Polyalphabetic Cipher

- Mono alphabetic cipher encoded using only one fixed alphabet.
- Polyalphabetic cipher is a substitution cipher in which the cipher alphabet for the plain alphabet may be different at different places during the encryption process.
    1. Vigenere cipher
    2. Vernam cipher



Polyalphabetic cipher [3]



Keyword : DECEPTIVE
Key : DECEPTIVEDECEPTIVEDECEPTIVE
Plaintext : WEAREDISCOVEREDSAVEYOURSELF
Cipher text: ZICVTWQNGRZGVTWAVZHCQYGLMGJ

An analyst looking at only the ciphertext would detect the repeated sequences VTW at a displacement of 9 and make the assumption that the keyword is either three or nine letters in length.

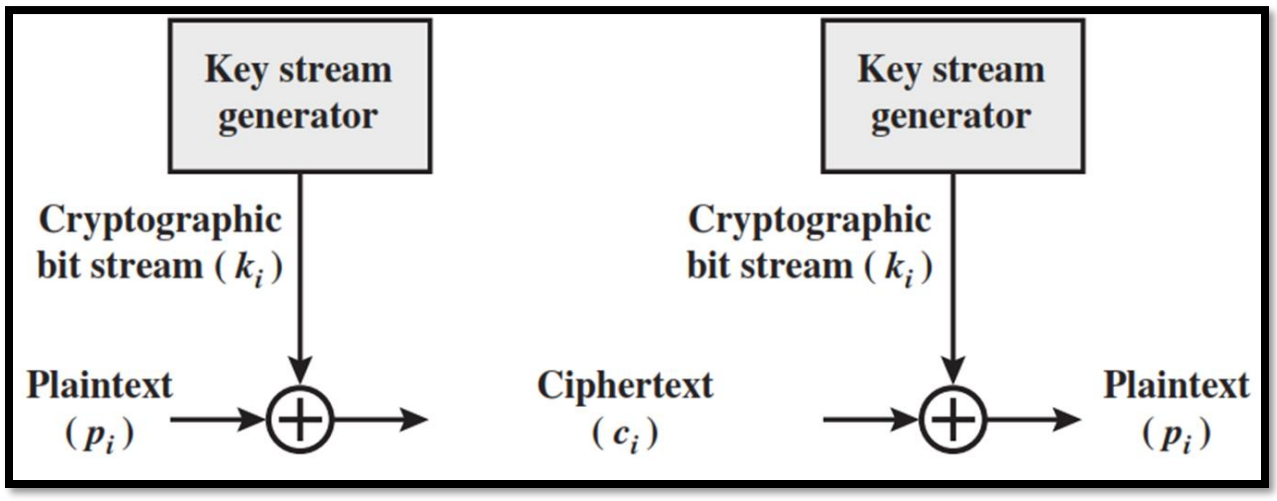
$$C = (P_1 + K_1, P_2 + K_2, \ldots P_m + K_m) \bmod 26$$

$$P = (C_1 - K_1, C_2 - K_2, \ldots C_m - K_m) \bmod 26$$

Keyword : DECEPTIVE
Key : DECEPTIVEWEAREDISCOVEREDSAV
Plaintext : WEAREDISCOVEREDSAVEYOURSELF

Vernam Cipher

- The cipher text is generated by applying the logical XOR operation to the individual bits of plaintext and the key stream.

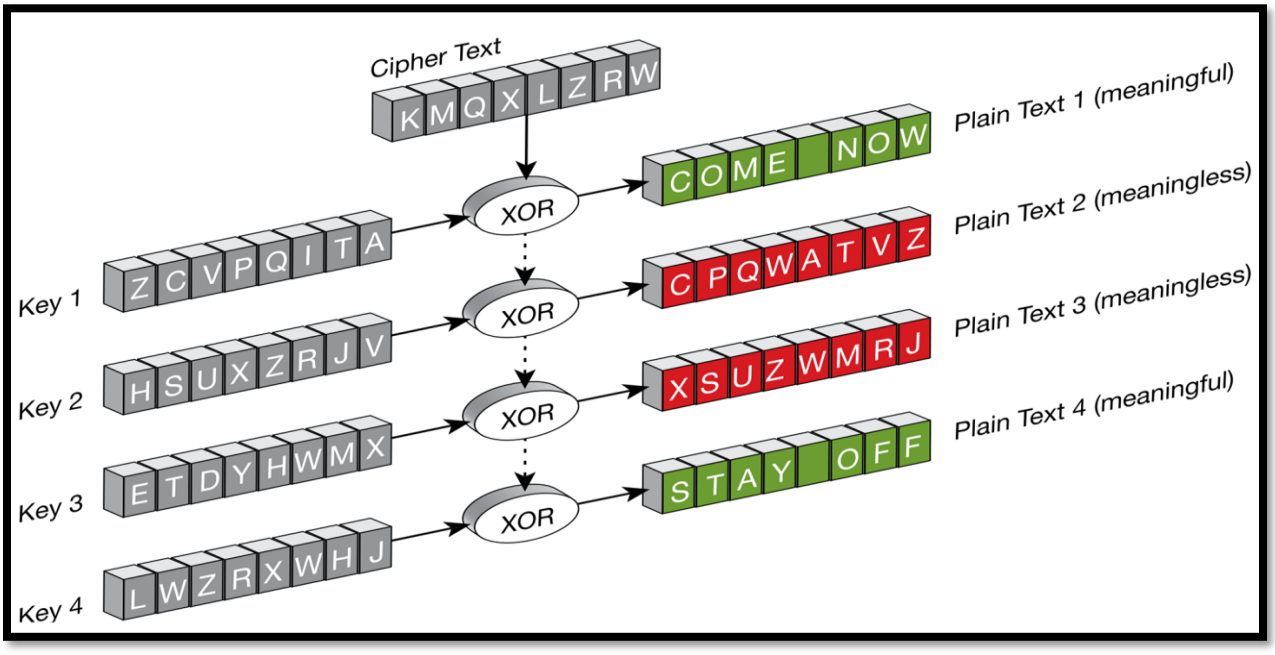


Vernam cipher [3]

One time pad

- The one-time pad, which is a provably secure cryptosystem, was developed by Gilbert Vernam in 1918.
- The message is represented as a binary string (a sequence of 0's and 1's using a coding mechanism such as ASCII coding.
- The key is a truly random sequence of 0's and 1's of the same length as the message.

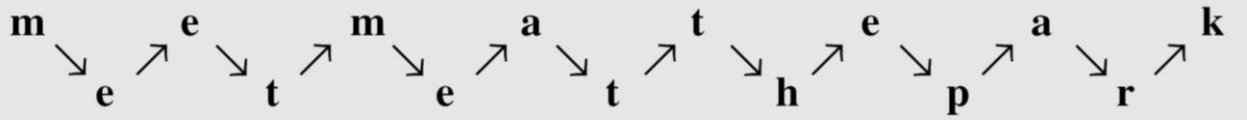
- message ='IF'
- then its ASCII code =(1001001 1000110)
- key = (1010110 0110001)
- *Encryption:*
  - 1001001 1000110 plaintext
  - 1010110 0110001 key
  - 0011111 1110110 cipher text



One time pad [3]

2.3 Transposition Techniques

- A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.
- The simplest such cipher is the rail fence technique, in which the plaintext is written down as a sequence of diagonals and then read off as a sequence of rows.
- For example, to send the message “Meet me at the park” to Bob, Alice writes



Transposition Techniques[3]

- She then creates the cipher text “MEMATEAKETETHPR”.

Rail fence technique

- A more complex scheme is to write the message in a rectangle, row by row, and read the message off, column by column, but permute the order of the columns.
- The order of the columns then becomes the key to the algorithm.

Key: 4 3 1 2 5 6 7
1 2 3 4 5 6 7
Plaintext: a t t a c k p
o s t p o n e
d u n t i l t
w o a m x y z
Cipher text: TTNAAPTMTSUOAODWCOIXKNLYPETZ

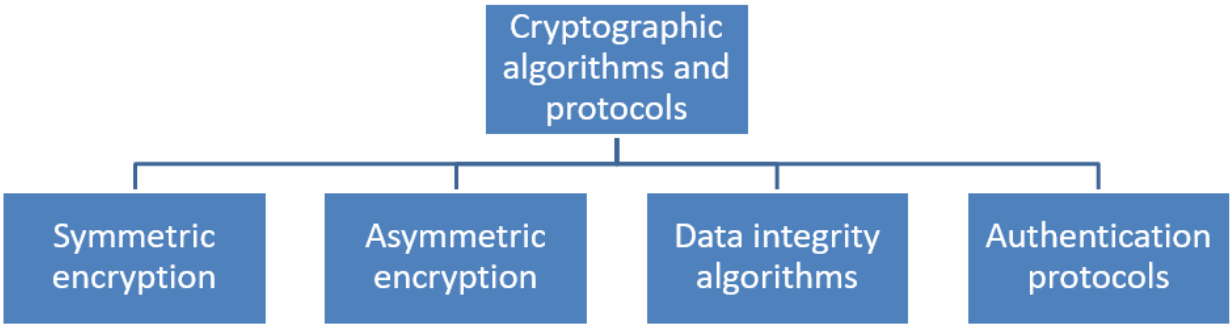
Cryptography and Cryptanalysis

- Cryptography and Cryptanalysis
  - Cryptography is the study of the design of techniques for ensuring the secrecy and/or authenticity of information
  - Cryptanalysis deals with the defeating such techniques to recover information, or forging information that will be accepted as authentic



Cryptographic Algorithms

- Cryptographic algorithms and protocols can be grouped into four main areas



Cryptographic Algorithms [3]

- Symmetric encryption used to secure the contents of blocks or streams of data of any size, including messages, files, encryption keys, and passwords
- Asymmetric encryption used to conceal small blocks of data, such as encryption keys and hash function values, which are used in digital signatures.
- Data integrity algorithms used to protect blocks of data, such as messages, from alteration
- Authentication Protocols are schemes based on the use of cryptographic algorithms designed to authenticate the identity of entities.

Security Objectives

- Security objectives for information and computing services are Confidentiality, Integrity, Availability, Authenticity, and Accountability.

1) Confidentiality:
    - Data confidentiality: Assures that private or confidential information is not made available or disclosed to unauthorized individuals.
    - Privacy: Assures that individuals control what information related to them may be collected and stored and by whom and to whom that information may be disclosed.



2) Integrity:
    - Data integrity: Assures that information and programs are changed only in a specified and authorized manner.
    - System integrity: Assures that a system performs its intended function in an unimpaired manner, free from deliberate or inadvertent unauthorized manipulation of the system.

3) Availability:
    - Assures that systems work promptly and service is not denied to authorize users.

4) Authenticity:
    - The property of being genuine and being able to be verified and trusted; confidence in the validity of a transmission, a message, or message originator.
    - This means verifying that each input arriving at the system came from a trusted source.

5) Accountability:
    - The security goal that generates the requirement for actions of an entity to be traced uniquely to that entity.
    - This supports nonrepudiation, deterrence, fault isolation, intrusion detection and prevention, and after-action recovery and legal action.

Threat and Attack

- Threat: A potential for violation of security, which exists when there is a circumstance, capability, action, or event that could crack security and cause harm. That is, a threat is a possible danger that might exploit a vulnerability.

- Attack: A violation on system security that derives from an intelligent threat; that is, an intelligent act that is a calculated attempt to avoid security services and violate the security policy of a system.



2.4 Steganography

- Steganography is the art and science of writing hidden messages in such a way that no one apart from the sender & intended recipient knows of the existence of the message.

- The word steganography is of Greek origin words steganos (στεγανός), meaning "covered or protected", and graphei (γραφή) meaning "writing".

- Cryptography relies on transformation algorithms using key to scramble a message.

- Steganography takes one piece of information and hides it within another.

- Both are used to protect information but steganography is concerned with hiding information thereby making it unseen while cryptography is concerned with encrypting information thereby making it unreadable.

- Some Examples:

- **Character marking:** where selected letters of text are overwritten with a special pencil. The marks are ordinarily not visible unless the paper is held at an angle to bright light.

- **Use of invisible ink**. Visible when some heat or chemical applied

- **Pin punctures.** On selected letters visible in front of light

- Typewriter correction ribbon visible under strong light

Types of Steganography:

Audio steganography
Video steganography
Textual steganography
Image steganography



References

1. Information Security Articles, References, & Blogs, UC SANTA BARBARA
2. https://www.it.ucsb.edu/general-security-resources/information-security-articles-references-blogs
3. What is Information Security? cisco
4. https://www.cisco.com/c/en_in/products/security/what-is-information-security-infosec.html
5. CRYPTOGRAPHY AND NETWORK SECURITY. Harlow: Tata McGraw-Hill.
6. Nutt, G. J. (2004). Operating systems: A modern perspective. Boston: Pearson/Addison Wesley.
7. Cyber Security Strategies. Tutorials point.
8. https://www.tutorialspoint.com/information_security_cyber_law/cyber_security_strategies.htm